VISHAL SELVAN

Senior React.js Developer | Front-End Engineer

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PROFESSIONAL SUMMARY

Front-End Engineer with 3.7+ years building scalable React.js applications in TypeScript and JavaScript (ES6+). Deep expertise in React core principles — hooks, component lifecycle, state management, and context — paired with hands-on experience in Next.js SSR/SSG, Redux Toolkit, React Router, and Micro-Frontend architectures. Experienced in **integrating AI-powered features** including **streaming LLM APIs (Claude, GPT-4, Gemini)**, AI chatbots with file attachment handling, MCP server orchestration, and smart UI components. Delivered 30%+ performance improvements across production dashboards through Webpack/Babel optimisation, lazy loading, code-splitting, and Lighthouse auditing. Proficient in RESTful API design, JWT-based auth, Axios interceptors, Jest unit testing, WCAG 2.1 AA accessibility, and CI/CD pipelines (Jenkins, Docker, AWS). Strong communicator with a track record of translating technical decisions for cross-functional stakeholders.

TECHNICAL SKILLS

Languages	TypeScript (Expert), JavaScript (ES6+), HTML5, CSS3, SCSS
Frameworks & Libraries	React.js (Hooks, Lifecycle, Context, Styled Components), Next.js (SSR/SSG), Redux Toolkit, MobX, React Router, Node.js, Material-UI, PrimeReact, Tailwind CSS, Angular (familiar)
AI & LLM Integration	Claude API (Anthropic), OpenAI GPT-4 API, Gemini API, Streaming LLM Responses (SSE / ReadableStream), AI Chatbot UI, Prompt Engineering, MCP (Model Context Protocol) Server Orchestration, Multimodal API Payloads, LangChain (familiar)
JavaScript Concepts	Promises & Async/Await, Prototypal Inheritance, Object & Array Manipulation, Closures, Event Loop
Build & Tooling	Webpack (code-splitting, plugins, loaders), Babel, Vite, ESLint, NPM
Architecture	Micro-Frontend (Module Federation), Component-Based Design, OOJS, Web App Performance Optimisation
Web APIs	RESTful API Design, WebSocket, Server-Sent Events (SSE), Web Storage, JWT, Axios, API Interceptors
Testing & QA	Jest, Unit Test Cases, Chrome DevTools, Lighthouse, Performance Auditing
DevOps & Cloud	Git (Gitflow), Docker, Kubernetes, Jenkins CI/CD, AWS (Solutions Architect Associate)
Data Visualisation	Recharts, PrimeReact DataTable
UX & Accessibility	WCAG 2.1 AA, Mobile-First Responsive Design, Cross-Browser Compatibility

PROFESSIONAL EXPERIENCE

Senior Software Engineer | LTIMindtree Coimbatore, India | July 2022 – Present

Cloud Cost Optimizer — React.js + TypeScript SPA | AI Chat Assistant | 1,500+ Recommendations

- Architected a scalable React.js + TypeScript SPA using React core principles — hooks (useState, useEffect, useMemo), component lifecycle, and Context API — handling 1,500+ cloud cost recommendations via PrimeReact DataTable, reducing manual analysis time by 40%
- Engineered an **AI Chat Assistant** by integrating streaming APIs from **Claude (Anthropic), GPT-4 (OpenAI), and Gemini (Google)** — enabling users to query cloud cost data conversationally with token-by-token response rendering using **Server-Sent Events (SSE) and fetch ReadableStream** for smooth, real-time AI output
- Built a **multi-model & MCP server selection interface** allowing users to switch between Claude, GPT-4, and Gemini, and select from configured **MCP (Model Context Protocol) servers** for context-aware, environment-specific AI responses

- Implemented **file attachment support** in the AI chat UI — accepting images and documents, encoding multimodal payloads for LLM APIs, and rendering image attachments inline with **interactive zoom and one-click download functionality**
- Built real-time data visualisations using Recharts with React composition patterns, memoisation, and virtual rendering — accelerating stakeholder decision-making by 25%
- Integrated RESTful APIs with TypeScript-typed request/response models, Axios interceptors, error handling, and data validation middleware — improving system reliability by 30%
- Implemented Redux Toolkit for centralised state management with normalised data structures and TypeScript interfaces, reducing defects by 15% and improving maintainability
- Optimised bundle performance using Webpack code-splitting, Babel transpilation, dynamic imports, and React.memo — boosting Lighthouse score by 20 points to 92 with measurable LCP and TTI improvements
- Authored Jest unit test cases for critical UI components and business logic, and presented architecture decisions to cross-functional stakeholders

Policy Compliance Dashboard — Next.js SSR/SPA | AWS Policy Tracking | 1,000+ Users | RBAC

- Designed a Next.js SSR SPA leveraging SSR/SSG for faster FCP, SEO-ready rendering, and reduced client-side JS payload — reducing re-rendering overhead by 20% and improving audit accuracy by 30%
- Applied advanced JavaScript (ES6+) patterns — Promises, prototypal inheritance, object/array manipulation — using OOJS principles and component-based architecture for clean, maintainable code
- Implemented Okta RBAC authentication with JWT token management, Axios API interceptors, and TypeScript-typed role definitions for 3 distinct user roles
- Leveraged Web Storage (sessionStorage) for client-side caching of policy state, reducing redundant API calls and improving perceived performance
- Applied WCAG 2.1 AA accessibility standards and mobile-first responsive design using HTML5, CSS3, SCSS, and Tailwind CSS — ensuring full cross-browser and cross-device compatibility
- Executed performance audits with Chrome DevTools and Lighthouse; remediated via code-splitting and lazy loading — improving score from 68 to 92 and reducing load time by 25%
- Designed and managed deployment pipelines using Jenkins CI/CD, Gitflow, Docker, and AWS infrastructure — reducing release cycle time by 20%

Service Analytics Dashboard — Real-Time Monitoring | Service Health & Performance Metrics

- Built responsive React.js + TypeScript dashboard using Material-UI, Tailwind CSS, SCSS, and HTML5 with mobile-first UI/UX best practices across all devices and screen sizes
- Engineered Micro-Frontend architecture using Webpack Module Federation — enabling independent deployment of feature modules and reducing cross-team coupling
- Developed modular Recharts visualisation components with custom Hooks, React composition patterns, and efficient data aggregation algorithms for real-time service health metrics
- Implemented WebSocket integration for live data streaming, combined with Web Storage caching and React Router navigation for seamless real-time user experience
- Integrated RESTful APIs with TypeScript interfaces, retry logic, and error boundaries — achieving 99.9% uptime SLA compliance
- Managed application state with Context API and custom Hooks for efficient client-side data filtering and caching
- Configured Docker containerisation and Kubernetes orchestration for AWS deployment with optimised multi-stage Dockerfiles — reducing deployment time by 60%
- Adopted Gitflow branching strategy — enabling parallel feature development, hotfix management, and clean release cadences

CERTIFICATIONS & EDUCATION

AWS Certified Solutions Architect – Associate December 2023

AWS Certified Cloud Practitioner October 2022

Bachelor of Engineering — Automobile Engineering | CGPA: 8.20
Madras Institute of Technology, Anna University, Chennai | 2018 – 2022